SPMNet: A Siamese Pyramid Mamba Network for Very-High-Resolution Remote Sensing Change Detection

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***Abstract*— Very-high-resolution (VHR) remote sensing images are characterized by extremely high-spatial resolution, incorporating higher pixel density and larger image sizes, which pose challenges for existing methods to extract complex texture features. Furthermore, due to the wide-area and high-resolution imaging strategy, VHR change detection images suffer from a severe imbalance between change pixels and nonchange pixels, increasing the difficulty of handling change detection tasks. To address these challenges, we introduced the omnidirectional selective scan module (OSSM), which has the capability to process long sequences. By integrating it with the lightweight Siamese feature pyramid network (SFPN), we designed a hybrid CNN–Mamba backbone, referred to as Siamese pyramid Mamba (SPMamba). This backbone captures both global and local information within bitemporal feature maps at each stage, enhancing the precision of texture feature extraction. In addition, to integrate the semantic features from each branch in SPMamba and reduce noise interference from nontarget change areas, we developed a hybrid fusion module (HFM). The HFM consists of two fusion modules: the high- and low-channel fusion module (HLM) and the bilateral channel fusion module (BCM), which facilitates both feature-level and channel-level integration, enhancing the sensitivity of the model to subtle changes. Extensive experimental results demonstrate that SPMNet achieves the highest $F1$-score of 91.80%, 90.99%, and 96.04% on the WHU-CD, LEVIR-CD, and CDD-CD datasets, respectively, outperforming 11 state-of-the-art methods. Moreover, the effects of varying image sizes on model training are thoroughly analyzed.**

***Index Terms*— Change detection, hybrid fusion module (HFM), Siamese pyramid Mamba (SPMamba), very-high-resolution (VHR).**

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I. Introduction

THE dynamic characteristics of the surface of Earth are shown by various anthropogenic and natural changes, leading to continuous alterations [1]. Remote sensing change detection is the process of identifying and quantifying changes in surface features by comparing and analyzing remote sensing images acquired at different times. With the advancement of remote sensing image acquisition technologies, very-high-resolution (VHR) remote sensing image change detection has emerged as a fundamental task in image interpretation, providing deeper insights and more comprehensive information for a range of real-world applications. VHR images play a crucial role in multiple fields, including land cover analysis [2], agricultural management [3], environmental monitoring [4], and disaster response [5]. Notably, the existing deep learning methods for remote sensing change detection can be broadly categorized into CNN-based and Transformer-based models. Fig. 1 illustrates the issues and challenges that current deep learning approaches face when processing VHR remote sensing images.

With the advancement of modern optical technologies, optimized optical systems ensure the precise capture of light signals and minimize distortions, thereby maintaining ultra-high clarity of lines and edges in VHR remote sensing images. In addition, high-precision sensors are capable of capturing more image texture details and multiscale spatial structures. In recent years, many early deep learning-based change detection methods [6], [7], [8], [9], [10], [11] predominantly employed Siamese CNN models, such as the classical ChangeNet [9] and the widely used U-Net [12] in remote sensing. Although CNNs have possess powerful visual feature extraction capabilities, their fixed-size convolutional kernels limit the receptive field, hindering the effective capture of global contextual information. This limitation results in the loss of crucial information within complex textures. Furthermore, in images where the background and foreground are imbalanced, the background typically occupies the majority of the pixels, while the foreground only occupies a small fraction. Consequently, CNN models are prone to learning features more related to the background, leading to the suboptimal detection and classification performance of foreground objects. This issue is particularly pronounced when the foreground objects are small, as the model may easily overlook or misclassify these objects.

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It is noteworthy that with the advent of attention mechanisms [13], Transformer models have been integrated into remote sensing visual tasks, demonstrating a strong capability to capture global information. In the context of remote sensing image change detection, systems typically acquire images of the same region from multiple angles and at different time points. Unlike natural images, VHR remote sensing images display diverse texture features, making it essential to model long-range context and extract large-scale spatial features from various directions [14]. For remote sensing images with complex textures, Transformer models, which lack explicit convolutional operations and layer-by-layer feature extraction, are prone to overlooking critical local texture details and multilevel features. Hence, the multiscale convolutional feature extraction is particularly important. Moreover, in change detection datasets with a significant imbalance between change and nonchange pixels, the self-attention mechanism of Transformer models tends to distribute attention evenly across the entire image, causing the model to focus excessively on background information and overlook subtle changes in the foreground, leading to decreased change detection performance. Furthermore, VHR remote sensing images usually contain high-precision numerical information, with each pixel storing rich texture and feature details, resulting in a quadratic increase in computational complexity for the attention mechanism. This significantly limits their performance in dense detection and real-time applications involving VHR remote sensing data.

Recently, state-space model (SSM) with hardware-efficient design, such as the Mamba deep learning model [15], has shown considerable promise in modeling long sequences. Inspired by Mamba, subsequent models such as Vim [16], VMamba [17], and RSMamba [18] have adopted SSM, continuously optimizing the scanning mechanism to extract rich spatial features and achieve globally effective receptive fields in image processing. However, due to significant differences between remote sensing images and natural images—such as the top–down view and much higher spatial resolution of VHR remote sensing images. Vim and VMamba are not fully optimized for VHR remote sensing image change detection. To address these limitations, the RSMamba model, an enhancement of VMamba, introduces a novel omnidirectional selective scan module (OSSM). OSSM is a novel feature extraction unit specifically designed for semantic segmentation and change detection tasks in VHR remote sensing images. Compared with previous SSM models, OSSM scans the input image patches in horizontal, vertical, diagonal, and their respective reverse directions, and then stacks these scanned image patches along new dimensions before feeding them into the SSM for global modeling, thereby achieving omnidirectional extraction of image texture features. Although RSMamba demonstrates promise in capturing spatial feature differences, its initial design as a general-purpose model for remote sensing images limits its effectiveness in VHR remote sensing image change detection.

In response to the challenges identified, we design a Siamese feature pyramid network (SFPN) with shared parameters, building on the traditional feature pyramid network (FPN) [19]. The SFPN extracts image features layer by layer using a bottom–up approach and connects multiple Mamba-based encoders laterally to construct a change detection network backbone for multiscale image texture feature extraction, referred to as Siamese pyramid Mamba (SPMamba). Specifically, SPMamba performs multiscale downsampling on the original image and inputs local detail information into Mamba to compute multilevel spatial feature differences. This approach facilitates the integration of global and local information, enabling the model to effectively learn both macrolevel overall texture structures and microlevel detailed textures. As a result, the model's ability to recognize positive samples among a large number of negative samples is significantly improved. Furthermore, considering the large spatial extent often present in VHR bitemporal remote sensing images, we also propose hybrid fusion module (HFM). The HFM consists of the high- and low-channel fusion module (HLM) and the bilateral channel fusion module (BCM). HLM is responsible for fusing semantic features between adjacent scales of the same image, allowing the model to capture rich spatial hierarchical information while reducing background noise interference. In addition, BCM performs feature-level fusion of semantic features at the same scale to enhance the representation of the image foreground. The main contributions of this article can be summarized as follows.

1) To address the complex texture structure of VHR remote sensing images and the substantial computational load in traditional models, we have designed a Siamese pyramid network backbone for change detection, termed SPMamba, which integrates SFPN and OSSM.
2) To further integrate multistage semantic features and reduce background noise interference, we proposed an HFM that performs both channel-level and feature-level fusion.
3) Extensive experiments and ablation studies are conducted on four widely recognized challenging remote sensing image change detection datasets (WHU-CD [20], LEVIR-CD [21], LEVIR-CD+ [21], and CDD-CD [22] datasets) to validate the effectiveness and robustness of the proposed method.

The rest is organized as follows. In Section II, a brief review of related work on remote sensing change detection within the deep learning domain is presented. Section III provides a comprehensive description of the proposed method. In Section IV, extensive experimental results and visual comparisons are presented. Finally, the effectiveness and prospects of the methodology are concluded in Section V.

II. Related Works

In this section, we briefly introduce deep learning-based change detection methods, the development of Mamba, and the application of vision Mamba in remote sensing change detection.

A. Change Detection Based on Deep Learning

The advancement of deep learning technologies has significantly enhanced the application of CNN models in

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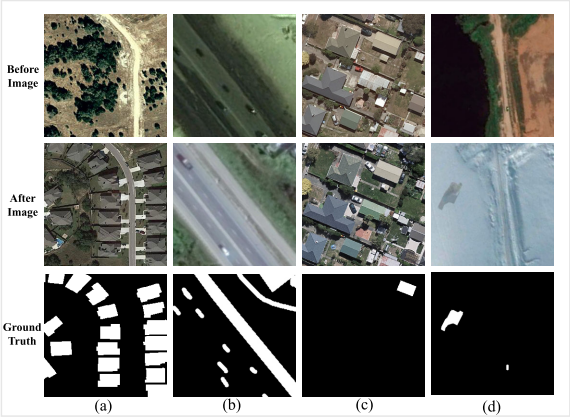


Fig. 1. Illustration of challenging scenarios, such as scenes with rich details and extensive background areas. (a) and (b) Variety of different texture patterns. (c) and (d) Imbalance between changed and unchanged areas.

change detection tasks. Siamese CNN models are particularly effective for this purpose, as they extract features from bitemporal remote sensing images using convolutional neural networks with shared parameters and then compare these features to identify regions of change and no change. In the context of 2-D remote sensing image change detection, Daudt et al. [23] proposed two Siamese extensions of fully convolutional networks, which achieved state-of-the-art performance on the then-existing change detection datasets. Furthermore, Zhang et al. [24] introduced a framework for detecting changes in 3-D laser scanning point clouds across different periods. These studies demonstrate that Siamese CNN-based networks can effectively handle a wide range of remote sensing image change detection challenges.

However, VHR remote sensing images, characterized by their complex texture features, present significant challenges for traditional CNN models. In contrast, Transformer models [21], [25], [26], [27], [28], which are built around attention mechanisms [13], model the entire image to extract global information and dependencies, ushering in a new era for remote sensing image processing. Chen et al. [29] proposed BIT to efficiently and effectively model context in the spatiotemporal domain. Despite these advancements, Transformer-based models often demand substantial computational resources due to their quadratic complexity when processing VHR remote sensing images. Consequently, there is an urgent need for lightweight CNN–Transformer hybrid models for dense prediction tasks in remote sensing. Li et al. [30] introduced an end-to-end encoder–decoder hybrid Transformer model that employs CNN to extract feature maps from input images and captures long-range feature dependencies through the Transformer. While these methods improve detection accuracy, significant potential for further enhancement and optimization remains.

B. Mamba in Deep Learning

In recent years, SSMs have made significant advancements in the field of deep learning, particularly in handling long-range dependencies and sequence data. Originally applied in control systems, SSM focuses on constructing state representations that integrate input sequences to predict output sequences. With the development of deep learning technologies, SSM has evolved from initially using zero-order hold (ZOH) techniques to discretize continuous signals and function mapping into sequence mapping. The introduction of the HiPPO [31] initialization method has further enhanced the model's capability to manage long-range dependencies. Notable models, such as LSSL [32] and S4 [33], have addressed issues related to computational and memory efficiency, with S4 reducing computational overhead through normalization and parameterization techniques. However, the state matrix in S4 does not provide targeted inferences based on varying sequence inputs. Advancements in research have led to various SSM variants that improve structural and efficiency aspects, including complex diagonal structures, multiple-input–multiple-output configurations, and diagonal plus low-rank structures [34], [35]. The introduction of Mamba [15] marks a transition to the S6 stage of SSM, showcasing performance that surpasses Transformers in natural language processing through selective scanning of input information and the implementation of novel hardware-aware algorithms.

Similar to Transformers, Mamba was initially designed for natural language processing and has shown efficiency in handling long-time sequences. Its scalability suggests that Mamba could be a promising alternative to Transformers in language modeling. However, a general backbone network based solely on SSM for handling visual data, such as images and videos, has not been extensively explored. Consequently, Zhu et al. [16] introduced Vim, which treats image patches as sequence data in Vim blocks, effectively compressing visual representations through a bidirectional selective state-space approach. Concurrently, Liu et al. [17] proposed VMamba, a model for visual tasks that improves parallel selective scanning algorithms and introduces 2-D selective scan (SS2D) specifically designed for image processing, achieving a four-directional scanning mechanism in the spatial domain. The application of visual Mamba has shown encouraging results in natural vision tasks.

C. Visual Mamba in Remote Sensing Change Detection

In the field of remote sensing image processing, Mamba is gradually replacing Transformers as the backbone network for various remote sensing tasks. Vim [16] and VMamba [17] leverage positional encoding and scanning mechanisms to enhance the Mamba model's sensitivity to spatial positions in image data, providing an alternative approach to modeling long-range dependencies in vision tasks. Although Mamba achieves linear complexity and high detection accuracy when processing VHR remote sensing images, existing models are often structurally simplistic and struggle to handle more challenging change detection tasks.

Currently, vision Mamba-based remote sensing change detection methods can be categorized into two main approaches: external integration of CNNs with vision Mamba and internal modifications to the vision Mamba module. In the first category, Paranjape et al. [36] proposed an encoder

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and difference module based on VMamba, enhancing the model's ability to capture channel and spatial information. To improve Mamba's capability in complex change detection scenarios, Zhang et al. [37] introduced edge-feature enhancement and the dual-flow state-space mechanism. In addition, Liu et al. [38] combined Mamba blocks with CNN-based feature extraction modules to capture both global and local features, thereby improving the model's representational capacity. However, these approaches often overlook the limitations of unidirectional and bidirectional scanning mechanisms in high-resolution settings.

Regarding internal modifications to the vision Mamba module, Zhang et al. [39] proposed a model called CDMamba, which integrates convolutional operations within Mamba to alleviate its lack of fine-grained details and improve precision in dense prediction tasks. Zhao et al. [18] introduced OSSM to further refine the selective scanning strategy for segmentation and detection in VHR remote sensing images.

Overall, most vision Mamba-based remote sensing change detection methods adopt a CNN–Mamba hybrid structure, utilizing SSM processing units to extract complex image features and achieve high-precision local and global change detection. Our goal is to combine these two improvement approaches by integrating CNNs with Mamba for multiscale feature extraction while leveraging the optimized OSSM mechanism to enhance Mamba's sequence modeling capability, enabling the model to better perceive both global and local information.

III. Methodology

In this section, we begin by introducing the core mechanism of the OSSM, followed by a comprehensive overview of the overall architecture of SPMNet.

A. Omnidirectional Selective Scan Module

In the field of computer vision, OSSM [18] refers to a class of structured SSM [33] derivative models, such as the Transformer [13] and variant Swin Transformer [40]. The core of OSSM is based on the control theory SSM. SSM defines a latent state $h(t)$ that represents the transformation from an input sequence $x(t)$ to an output sequence $y(t)$. Its principle can be represented by linear ordinary differential equation (ODE) as follows:

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t) \tag{2}$$

where $\mathbf{A} \in \mathcal{R}^{N\times N}$ serves as the state-space update matrix of the model, while $\mathbf{B} \in \mathcal{R}^{N\times 1}$ and $\mathbf{C} \in \mathcal{R}^{1\times N}$ represent the sequence mapping weights.

To handle discretized data (text sequences), SSM treats the sequence as samples from a continuous signal. ZOH technique is a technique used in digital control systems to implement signal sampling and holding. SSM utilizes ZOH to define new learnable parameters Δ and applies ZOH operations to $\mathbf{A}$ and $\mathbf{B}$. The discretization can be represented as follows:

$$\overline{\mathbf{A}} = \exp(\Delta\mathbf{A}) \tag{3}$$

$$\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{A} \tag{4}$$

where Δ represents the input stage holding and $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ are the results of the discretization of parameters $\mathbf{A}$ and $\mathbf{B}$.

After that, the discretization of (1) and (2) can be formulated as follows:

$$h'_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \tag{5}$$

$$y_t = \mathbf{C}h_t. \tag{6}$$

Finally, to effectively address the long-range dependency issues in sequence modeling within limited storage space, HiPPO [31] is introduced to construct the hidden state matrix $\mathbf{A}$ as follows:

$$\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^* - \mathbf{P}\mathbf{Q}^{\mathbf{T}} = \mathbf{V}\left(\Lambda - \left(\mathbf{V}^*\mathbf{P}\right)\left(\mathbf{V}^*\mathbf{P}\right)^*\right)\mathbf{V}^* \tag{7}$$

where Λ denotes the diagonalization operation and $\mathbf{V} \in \mathcal{C}^{N\times N}$ is the unitary matrix for diagonalizing $\mathbf{A}$. $\mathbf{P}$, $\mathbf{Q} \in \mathcal{R}^{N\times r}$ represent the low-rank decompositions.

Compared with SSM used in natural language processing, OSSM integrates an omnidirectional scanning mechanism, facilitating the conversion of image patches in the visual tasks into sequences that can be processed by SSM. OSSM performs omnidirectional scanning of the input image patches in horizontal, vertical, diagonal, antidiagonal, and their respective reverse directions. The resulting image patch sequences are then stacked along new dimensions and fed into the SSM block, ultimately achieving omnidirectional spatial feature extraction. In addition, due to the global attention mechanism of OSSM, it can capture local foreground features from the large background when handling the class imbalance issue, preventing the model from overlooking change elements. Overall, OSSM is efficient and lightweight, allowing more learnable modules to be stacked under a similar total depth budget when building the model.

B. Overall Architecture

As shown in Fig. 2(a), SPMNet is a hybrid Siamese network architecture based on the integration of CNN and Mamba. It primarily consists of two components: the SPMamba backbone with shared weights and the HFM. The SPMamba backbone is designed to extract global contextual spatial information and learn the complex texture structures of images, while the HFM facilitates the fusion of multiscale high-level semantic information and minimizes noise interference from imbalanced change pixels.

In summary, SPMNet is designed as a comprehensive and complex SFPN architecture, which encompasses four stages of low-level feature extraction and high-level semantic analysis. Leveraging the architectural principles of SFPN, SPMNet applies a bottom–up downsampling operation to the input images and feeds the resulting low-level features into an encoder. The encoder learns locally prominent image texture information and global spatial difference features, thereby achieving multiscale local and global feature extraction and analysis, which prevents the model from over-focusing on background information. The encoder primarily consists of depthwise separable convolutions (DSCs) and OSSM blocks, with OSSM referencing RSMCD [18]. Finally, the features extracted by the encoder are input into the corresponding

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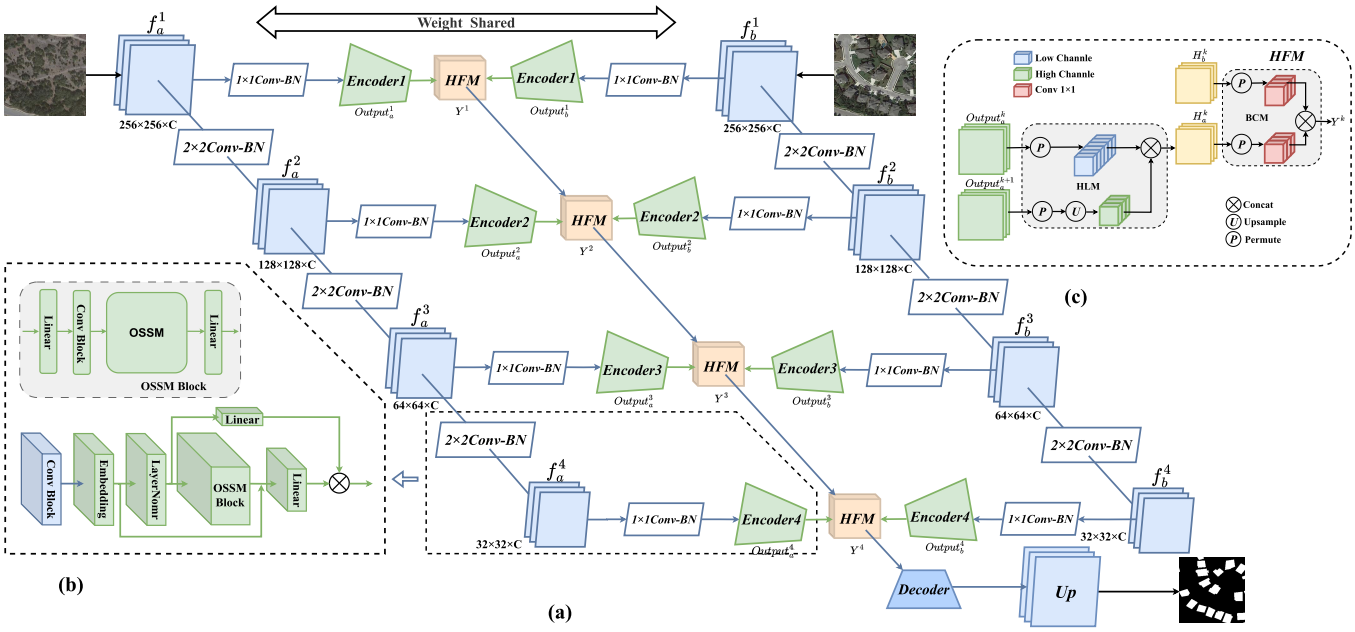


Fig. 2. Network architecture of the proposed SPMNet. (a) Implementation details of SPMNet. The multiscale features extracted by the Siamese pyramid are fed into the OSSM-based encoder for further extraction of spatial dimension features, where the model shares weights at this stage. (b) Single branch of SPMamba for downsampling and spatial feature extraction of feature maps. (c) HFM for multiscale semantic feature fusion, the HLM and BCM modules further remove interference noise.

scale fusion modules, HFM, to integrate multiscale high-level semantic features, which are further processed by the decoder to restore the original image size. This process effectively combines the shallow prominent features obtained by CNN with the multidirectional spatial information perceived by Mamba, thereby enhancing VHR remote sensing change detection from both global and local perspectives.

C. Siamese Pyramid Mamba

As shown in Fig. 2(a), SPMamba adopts a Siamese pyramid network architecture similar to TAEDet [41] and replaces the high-level semantic feature extractor of SFPN with an encoder centered around OSSM. In the SPMamba backbone, the backbone network processes bitemporal images (I_1, I_2), using unified network layers and parameters. The two branches share a single set of weights, meaning that the weights are represented in the same feature space. Therefore, the results of the overall weight-shared SPMamba backbone network can be expressed as follows:

$$\text{Output}_a = \text{SPMamba}(I_1), \quad \text{Output}_b = \text{SPMamba}(I_2). \tag{8}$$

Specifically, the single-path top–down downsampling in SPMamba comprises four stages, each extracting feature maps at different scales, which can be described as $\{f_i^m | i = a, b, m = 1, 2, 3, 4\}$. Here, f_i^m denotes the feature maps at each stage, i represents the temporal points of the bitemporal images, and m represents the four different downsampling scales. The low-level features are then horizontally connected to the encoders through $\text{Conv}_{1\times1}$ blocks to further identify change region features. Therefore, the result of each SPMamba branch can be expressed as follows:

$$\text{Output}_i^m = \text{Encoder}_i^m\left(\text{Conv}_{1\times1}\left(\text{Down}_i^m\left(f_i^m\right)\right)\right) \tag{9}$$

where f_i^m represents the input features at different temporal points. Down_i^m denotes the downsampling operation at stage m. $\text{Conv}_{1\times1}$ refers to a convolutional block with a kernel size of 1. Encoder_i^m represents the encoder at stage m for branch i.

In addition, Fig. 2(b) illustrates a detailed depiction of the internal processing structure of a single-branch SPMamba. The core component of the SPMamba block is OSSM. Specifically, images at different scales first undergo feature extraction through two basic convolutional blocks. The features following the patch embedding (PE) layer are subjected to layer normalization to standardize the input data. The normalized data are subsequently subjected to linear transformation and fed into deep convolutional layers to reduce the number of parameters. The convolved results are further processed by OSSM to learn the implicit overall texture structure within the locally extracted feature maps. Finally, the SPMamba block adds the features output by OSSM to the normalized results, and after a linear transformation, performs a residual connection with the PE features, achieving both global and local feature extraction. Mathematically, the above process can be modeled as follows:

$$R^1 = E_m\left(\text{Conv}_{1\times1}\left(\text{Conv}_{2\times2}\left(f_i^m\right)\right)\right) \tag{10}$$

$$R^2 = \text{Linear}(L_N(R_1)) \tag{11}$$

$$R^3 = \text{Cat}\left(R^2, \text{Linear}\left(D_W\left(\text{OSSM}\left(R^2\right)\right)\right)\right) \tag{12}$$

$$\text{Output}_i^m = \text{Cat}\left(R^1, \text{Linear}\left(R^3\right)\right) \tag{13}$$

where E_m, L_N, and D_W represent the PE layer, layer normalization, and DSC, respectively. OSSM denotes omnidirectional selective scanning module for computing spatiotemporal feature differences and Cat refers to the concatenation operation $(\cdot)$. SPMamba inherits the comprehensive omnidirectional directional spatial scanning mechanism of OSSM while integrating the multiscale feature capture of SFPN. This

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combination enables multiscale fusion of global and local information, allowing the model to learn complex spatial features and texture information, thereby improving the accuracy of change detection.

D. Hybrid Fusion Module

As shown in Fig. 2(c), the proposed HFM consists of HLM with a hybrid convolutional structure and BCM with parallel convolution. HLM performs channel-level fusion of semantic feature maps from various stages of SPMamba, integrating features from adjacent scales with simultaneous time points. The process not only significantly reduces the redundancy between channels but also effectively integrates the semantic features of different scales, thereby improving the detection accuracy even in cases of imbalanced sample distribution. Subsequently, the fused features are then fed into BCM to further enhance semantic features and improve the model's sensitivity to change features. BCM achieves feature-level fusion by integrating high-level semantic information from both time points, capturing subtle change relationships, and further improving the detection accuracy.

The HLM consists of two convolutional blocks with different channel scaling: low-channel convolution (LC) and high-channel convolution (HC). For semantic feature maps extracted from different scales by SPMamba, LC retains more high-level texture features and edge details, while HC, with fewer output channels, focuses on more abstract semantic features. This design approach not only minimizes noise interference from lower level features but also significantly reduces computational load. The operation of HLM can be represented as follows:

$$H_a^k = \mathrm{Cat}\left(\mathrm{Up}_b^2\left(H_C^{\frac{1}{4}}\left(\mathrm{Output}_a^{k+1}\right)\right), L_C^{\frac{1}{2}}\left(\mathrm{Output}_a^k\right)\right) \tag{14}$$

where Output_a^k and Output_a^{k+1} denote the low-level and high-level semantic feature maps, respectively. $H_C^{(1/4)}$ and $L_C^{(1/2)}$ refer to HC and LC, where the number of output channels is reduced to (1/4) and (1/2) of the original, respectively. Up_b^2 denotes the bilinear interpolation upsampling operation.

The BCM includes two convolutional blocks, each containing operations such as $\mathrm{Conv}_{1\times1}$, batch normalization, and ReLU activation. To further improve the accuracy of change detection, BCM integrates high-level semantic features from the bitemporal images to achieve feature-level fusion. This process can be mathematically represented as follows:

$$Y_a^k = \mathrm{Cat}\left(\mathrm{Conv}^a\left(H_a^k\right), \mathrm{Conv}^b\left(H_b^k\right)\right) \tag{15}$$

where H_a^k and H_b^k represent the outputs after channel-level fusion by HLM and Y_a^k denotes the result of BCM. Conv^a and Conv^b refer to the convolutional blocks corresponding to the feature maps. By incorporating the HFM module, the model integrates semantic information from features at the same scale but from different temporal images extracted by SPMamba, enhancing its capability to understand spatiotemporal relationships within the images and thereby improving detection accuracy.

E. Loss Function

Existing change detection datasets often exhibit extreme imbalance between changed and unchanged elements. This class imbalance may cause the model to focus more on learning the abundant negative samples while neglecting the positive samples that contain complex texture features. Therefore, to address this class imbalance issue, the overall loss function we adopted consists of two components: binary cross-entropy (BCE) loss [42] and Dice loss [43]. The BCE loss L_{BCE} and Dice loss L_{Dice} are formulated as follows:

$$L_{\mathrm{BCE}}(y, \hat{y}) = -\left(y \cdot \log\left(\hat{y}\right) + (1-y)\log\left(1-\hat{y}\right)\right) \tag{16}$$

$$L_{\mathrm{Dice}}\left(y, \hat{y}\right) = 1 - \frac{2 \cdot y \cdot \hat{y}}{y + \hat{y}} \tag{17}$$

where y and $\hat{y}$ are the ground-truth label and corresponding predicted change map, respectively. The value of pixels in y is 0 or 1; yet, the value of pixels in $\hat{y}$ from 0 to 1, represents the probability of being predicted as changed or unchanged. The overall loss function is defined as follows:

$$L\left(y, \hat{y}\right) = \lambda \cdot L_{\mathrm{BCE}}\left(y, \hat{y}\right) + \Phi \cdot L_{\mathrm{Dice}}\left(y, \hat{y}\right) \tag{18}$$

where λ and Φ are parameters that balance L_{BCE} and L_{Dice}. In the combined loss function, L_{BCE} is used for global convergence, balancing the overall prediction between changed and unchanged regions, while L_{Dice} focus on addressing the class imbalance issue with fewer positive samples (changed regions) and improving the detection performance of changed regions. The appropriate weight parameters λ and Φ will be determined through subsequent experimental analysis.

IV. Experiment

In this section, we will first provide a detailed description of the change detection datasets used in the experiments, the experimental environment, and the evaluation metrics. We will then conduct quantitative and qualitative analyses of the dataset results and perform ablation studies on the proposed modules. Finally, we will investigate the impact of feature maps at each stage of the model and the VHR remote sensing image sizes on model training.

A. Datasets

The WHU-CD dataset [20] is a building change detection dataset proposed by Wuhan University. It comprises bitemporal VHR RGB remote sensing images of Christchurch, New Zealand, captured in 2012 and 2016. The dataset covers an area of 32 507 × 15 354 pixels with a spatial resolution of 0.3 m/pixel. For fair comparison with other methods, the original images were cropped into nonoverlapping patches of 256 × 256 pixels and then randomly split into training, validation, and testing sets, with 4880, 672, and 1392 patches, respectively, following a 7:1:2 ratio.

LEVIR-CD [21] and its enhanced version, LEVIR-CD+, are datasets specifically designed for building change detection. They contain 637 pairs and 985 pairs of 1024 × 1024 remote sensing images, respectively, with a spatial resolution of 0.5 m/pixel. Due to GPU memory limitations,

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the images were cropped into 256 × 256 nonoverlapping patches. Specifically, the LEVIR-CD dataset is divided into 7120/1024/2048 patches for training/validation/testing, while the LEVIR-CD+ dataset is divided into 10 192/5568 patches for training/testing.

The CDD-CD dataset was acquired by Lebedev et al. [22] using Google Earth (DigitalGlobe) to capture seasonal change remote sensing images of the same area. This dataset includes seven pairs of images with a size of 4725 × 2200 pixels and four pairs with a size of 1900 × 1000 pixels. The images encompass various types of changes, such as trees, cars, and large building structures, with spatial resolutions ranging from 3 to 100 cm/pixel. The dataset authors cropped these 11 image pairs into nonoverlapping patches of 256 × 256 pixels. The training, validation, and testing sets were divided into 10 000, 3000, and 3000 images, respectively, in line with other methods.

B. Implementation Details and Metrics

The proposed model and related algorithms were implemented using the PyTorch framework [44] and trained on an NVIDIA RTX 3090. Before training, the image data was augmented by applying random horizontal flips and random transpositions with a probability of 50%. The image data was loaded in batches of 16, and the AdamW optimizer was used in conjunction with a loss function that combines BCE loss [42] with the Dice coefficient loss [43]. The model was trained for 100 epochs on each dataset, with an initial learning rate starting at 0.001 and weight decay set to 0.001. The learning rate was decreased by a factor of 0.1 if the $F1$-score did not improve over five consecutive epochs.

To assess the effectiveness of the proposed model, four key metrics were employed: precision (Pre.), recall (Rec.), $F1$-score ($F1$), and IoU. The $F1$-score provides a comprehensive measure by balancing precision and recall, while IoU quantifies the overlap between the predicted outcomes and the actual ground-truth pixel areas. These metrics collectively reflect the performance of the proposed method across multiple dimensions. They are defined as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$F1 = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{22}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative, respectively. They range from 0 to 1, where higher values indicate better performance.

C. Comparison With State-of-the-Art Methods

To validate the effectiveness of the CNN–Mamba hybrid structure-based SPMNet, we compare it with several state-of-the-art deep learning methods. These include fully convolutional networks (FC-EF [23], FC-Siam-Diff [23], FC-Siam-Conv [23], DTCDSCN [45], and SNUNet-C32 [46]), Transformer-based models (BIT-18 [29], ChangeFormerV6 [47], HANet [48], and SEIFNet [49]), and Mamba-based models (ChangeMamba-S [50] and RSM-CD [18]). Notably, FC-EF was the first to propose a Siamese fully convolutional neural network for change detection. SEIFNet proposed a lightweight network for multiscale adaptive fusion by comprehensively considering temporal differences and multiscale features. ChangeMamba is the first to explore the potential of the Mamba architecture in remote sensing CD tasks and design MambaBCD for binary change detection (BCD). RSM-CD improved the scanning mechanism, proposing a network backbone for remote sensing image segmentation and change detection. These methods present effective solutions for various challenges in CD.

1) Quantitative Evaluation: Table I presents the performance comparison between the proposed model and the most advanced methods in recent years. From the quantitative results, on the WHU-CD dataset, SPMNet further improves $F1$-score and IoU compared with the latest SEIFNet and RSM-CD, achieving 91.80% and 84.84%, respectively. In addition, on the LEVIR-CD dataset, SPMNet also demonstrates a significant advantage in comprehensive metrics, achieving the highest recall (90.58%), $F1$-score (90.99%), and IoU (84.07%), while also showing strong competitiveness in precision. A reasonable explanation for this is that SPMNet better captures spatial features and understands complex image texture features.

Although the WHU-CD and LEVIR-CD datasets consist of VHR remote sensing images, making it challenging for existing deep learning theories to perform accurate detection, these datasets primarily focus on building changes. To further validate the superiority and robustness of SPMNet, extensive comparative experiments were conducted on the LEVIR-CD+ and CDD-CD datasets. Compared with the LEVIR-CD dataset, LEVIR-CD+ has a longer time span and covers a wider range of buildings. As shown in Table I, the $F1$-score of SPMNet is 83.09%, which is close to the optimal SEIFNet's 83.35%. Although our method did not achieve the best performance, considering all the metrics, SPMNet shows no significant weaknesses and can provide stable results in complex scenarios. The CDD-CD dataset not only includes various types of land cover changes but also contains a certain number of VHR remote sensing images with resolutions ranging from 0.03 to 1 cm/pixel. As shown in Table I, SPMNet achieved the highest Precision (96.63%), $F1$-score (96.04%), and IoU (92.52%) on the CDD-CD dataset, indicating that SPMNet can capture global contextual information and learn the implicit relationships between complex land cover change types.

2) Qualitative Comparison: Figs. 3–6 display the visual comparison of different methods across the four datasets. Figs. 3–6(a)–(h) represent some challenging samples selected from the testing set. To better illustrate the performance differences between the compared models, we have used different color-coded pixel annotations in all the visual comparison results, where red indicates false negatives and green indicates false positives. In addition, to further enhance the clarity of

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TABLE I

QUANTITATIVE EXPERIMENTAL RESULTS(%) ON THE WHU-CD, LEVIR-CD, LEVIR-CD+, AND CDD-CD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, BLUE, AND GREEN

Methods	WHU-CD Pre. / Rec. / F1 / IoU	LEVIR-CD Pre. / Rec. / F1 / IoU	LEVIR-CD+ Pre. / Rec. / F1 / IoU	CDD-CD Pre. / Rec. / F1 / IoU
FC-EF [23]	70.05 / 80.20 / 74.78 / 59.72	78.61 / 73.53 / 75.98 / 61.21	56.22 / 55.95 / 56.09 / 38.97	78.78 / 56.94 / 66.10 / 49.38
FC-Siam-Diff [23]	87.54 / 76.24 / 81.50 / 68.77	90.29 / 83.76 / 86.90 / 76.83	79.40 / 75.42 / 77.36 / 63.01	87.20 / 57.13 / 69.02 / 52.71
FC-Siam-Conv [23]	79.97 / 81.00 / 80.48 / 67.33	88.06 / 85.91 / 86.97 / 76.94	66.75 / 76.61 / 71.34 / 55.45	83.50 / 61.82 / 71.04 / 55.09
DTCDSCN [45]	94.03 / 80.23 / 86.59 / 76.34	90.65 / 89.27 / 89.96 / 81.74	**84.82** / 72.84 / 78.38 / 64.44	94.27 / 93.62 / 93.94 / 88.57
SNUNet-C32 [46]	78.22 / 74.36 / 76.24 / 61.60	90.83 / 88.63 / 89.72 / 81.36	74.23 / 67.74 / 70.84 / 54.84	95.49 / 94.75 / 95.12 / 90.70
BIT-18 [29]	86.64 / 81.48 / 83.98 / 72.39	89.24 / 89.37 / 89.30 / 80.68	80.14 / **84.24** / 82.14 / 69.69	86.38 / 90.75 / 88.51 / 79.30
ChangeFormerV6 [47]	90.63 / 82.44 / 86.34 / 75.96	**92.65** / 88.75 / 90.66 / 82.91	80.06 / 79.56 / 79.81 / 66.41	94.89 / 92.88 / 93.88 / 88.45
HANet [48]	88.30 / 88.01 / 88.16 / 78.82	91.21 / 89.36 / 90.28 / 82.27	79.70 / 75.53 / 77.56 / 63.34	92.86 / 85.87 / 89.23 / 80.55
SEIFNet [49]	94.42 / 87.21 / 90.67 / 82.94	92.26 / 89.09 / **90.84** / 82.89	82.65 / **84.07** / **83.35** / **71.46**	**95.96** / **95.89** / **95.93** / **92.16**
ChangeMamba-S [50]	**95.63** / 79.45 / 86.79 / 76.67	91.10 / **90.34** / 90.72 / 83.01	83.02 / 81.62 / 82.31 / 69.94	94.96 / 95.05 / 95.01 / 90.49
RSM-CD [18]	93.37 / **90.42** / **91.80** / **84.83**	**92.76** / 89.73 / 88.94 / **83.85**	83.44 / 80.86 / 82.13 / 69.68	95.90 / 94.79 / 95.34 / 91.09
SPMNet	**94.67** / **89.10** / **91.80** / **84.84**	92.12 / **90.58** / **90.99** / **84.07**	**83.59** / 82.60 / **83.09** / **71.01**	**96.63** / **95.60** / **96.04** / **92.52**

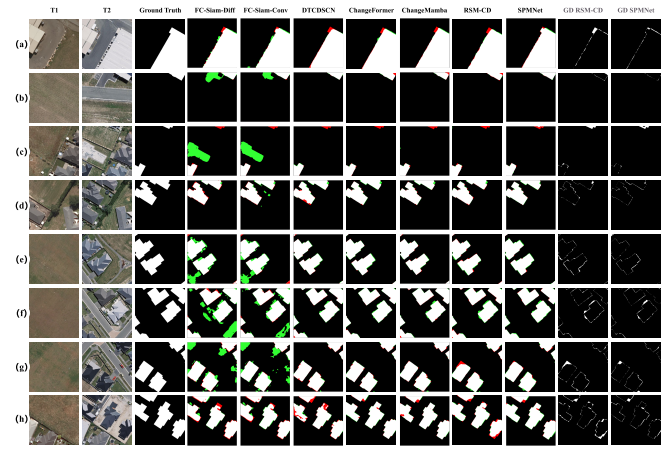


Fig. 3. Visual comparison with the proposed method and SOTA methods on WHU-CD datasets. (a)–(h) Eight representative samples in the WHU-CD dataset.

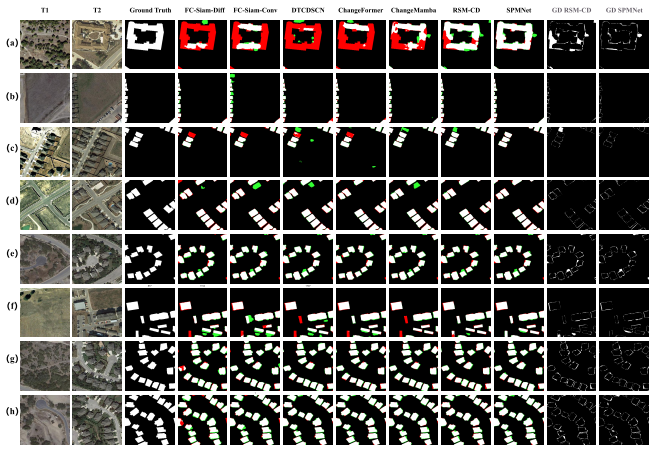


Fig. 4. Visual comparison with the proposed method and SOTA methods on LEVIR-CD Datasets. (a)–(h) Eight representative samples in the LEVIR-CD.

the visual comparison, we present grayscale difference maps between the results predicted by SPMNet and the Mamba-based RSM-CD model and the ground-truth labels.

a) Visualization on WHU-CD dataset: For the WHU-CD dataset, SPMNet exhibits fewer false positives in detection, especially in cases involving irregularly shaped buildings with complex texture structures [see Fig. 3(d)–(g)]. Specifically, in Fig. 3(a)–(c), for samples with highly imbalanced change and non-change pixel characteristics, FC-Siam-Diff and FC-Siam-Conv produce numerous false positives, while attention-based models such as ChangeFormerV6 suffer from missed detections. In addition, as shown in Fig. 3(h), compared with the Mamba-based models ChangMamba and RSM-CD, SPMNet generates fewer red pixels, indicating fewer missed detections. Further analysis of the grayscale difference maps (GD RSM-CD and GD SPMNet) reveals that SPMNet, leveraging the multiscale SPMamba blocks, demonstrates a more efficient feature extraction capability, enabling it to accurately capture irregularly shaped large buildings.

b) Visualization on LEVIR-CD: The LEVIR-CD dataset also primarily focuses on building changes. However, compared with the WHU-CD dataset, its image textures are more complex. In Fig. 4(a), (c), (f), (g), and (h), due to variations in lighting and vegetation, convolutional models (FC-Siam-Diff and FC-Siam-Conv) and Transformer-based models ChangeFormerV6 exhibit varying degrees of missed detections (marked in red pixels). In addition, in Fig. 4(b) and (d), due to the presence of small and densely distributed buildings, ChangMamba and RSM-CD produce false detections. As indicated by the grayscale difference maps (GD RSM-CD and GD SPMNet), SPMNet not only effectively captures global features but also more precisely identifies the edges of the change regions, resulting in detection outcomes that align more closely with the ground truth.

c) Visualization on LEVIR-CD+: Compared with LEVIR-CD dataset, the LEVIR-CD+ dataset not only covers a time span of 5–14 years but also includes the decline of newly constructed buildings such as villas, residential areas, and garages. As shown in Fig. 5(a) and (d), when the images before and after the change are highly similar, traditional CNN-based methods (FC-Siam-Conv and DTCDSCN) struggle to detect subtle change instances. As illustrated in Fig. 5(b), (c), and (h), for large-scale building changes, even standard Transformer and Mamba-based models designed

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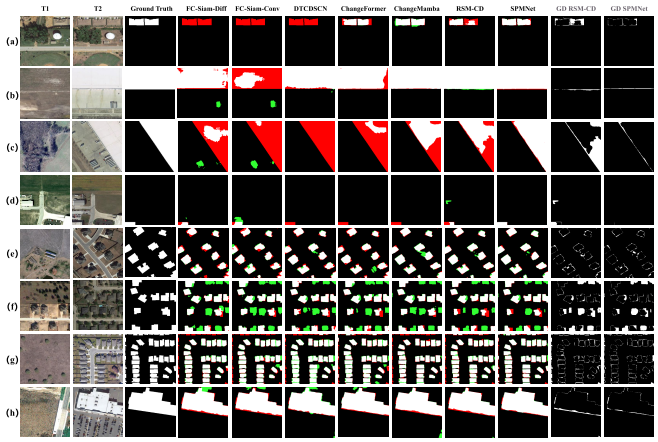


Fig. 5. Visual comparison with the proposed method and SOTA methods on LEVIR-CD+ Datasets. (a)–(h) Eight representative samples in the LEVIR-CD+.

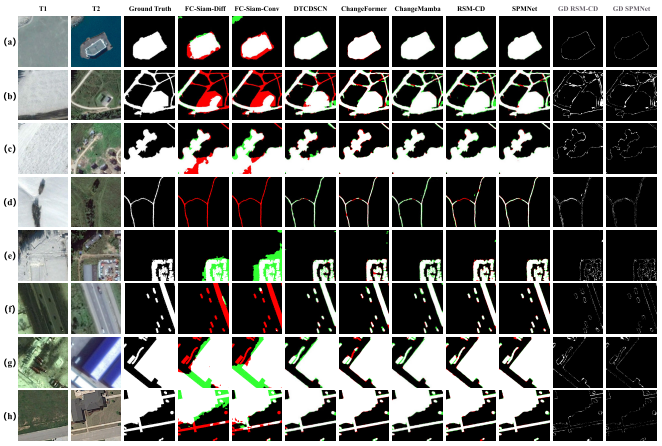


Fig. 6. Visual comparison with the proposed method and SOTA methods on CDD-CD Datasets. (a)–(h) Eight representative samples in the CDD-CD.

for capturing long-range dependencies (ChangeFormerV6 and ChangMamba) fall short. The comparison between GD RSM-CD and GD SPMNet reveals that SPMNet aligns more closely with the actual change patterns. In addition, for continuous small-object buildings, as shown in Fig. 5(e), (f), and (g), SPMNet demonstrates higher accuracy.

d) Visualization on CDD-CD: Compared with the WHU-CD and LEVIR-CD datasets, the CDD-CD dataset not only includes VHR remote sensing images but also involves more complex types of changes, such as seasonal variations, vegetation, and building changes. Fig. 6(a), (g), and (h) illustrates a combination of seasonal and building changes, while Fig. 6(b)–(d) primarily depicts the land and vegetation changes. These composite changes often make it challenging for models to identify change features or cause confusion in feature detection, ultimately leading to large-scale false positives and missed detections, as observed in FC-Siam-Diff, FC-Siam-Conv, and DTCDSCN. Furthermore, when the change elements involve vehicles, issues such as varying vehicle densities on roads, background interference (e.g., asphalt roads), and small-scale changes often arise. As shown in Fig. 6(e) and (f), SPMNet demonstrates higher accuracy in

TABLE II
COMPARISON OF COMPUTATIONAL COMPLEXITY ON THE CDD-CD DATASET

Methods	Params(M)	FLOPs(G)	F1(%)
FC-EF [23]	1.35	3.58	66.10
FC-Siam-Diff [23]	1.35	4.73	69.02
FC-Siam-Conc [23]	1.55	5.33	71.04
DTCDSCN [45]	31.26	13.21	93.94
SNUNet-C32 [46]	12.03	54.83	95.12
BIT-18 [29]	3.06	8.75	88.51
ChangeFormerV6 [47]	41.02	202.78	93.88
HANet [48]	2.61	17.62	89.23
SEIFNet [49]	27.91	8.37	95.93
ChangeMamba-S [50]	48.93	25.82	95.01
RSM-CD [18]	20.26	13.03	95.34
SPMNet(ours)	17.33	6.94	96.04

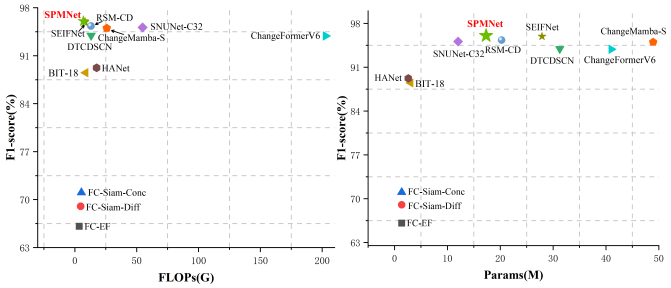


Fig. 7. Model complexity comparison of proposed method and SOTA methods on CDD-CD dataset.

predicting these changes compared with Mamba-based models (ChangeMamba and RSM-CD), with its grayscale difference maps aligning more closely with the ground truth. This further indicates that SPMNet possesses stronger semantic detection capabilities, allowing it to better distinguish between background and foreground change regions.

3) Parameter Comparison: To evaluate the effectiveness of the models, as shown in Table II, we analyzed and compared them in terms of the number of parameters (Params), floating-point operations (FLOPs) and $F1$-score. The analysis reveals that SPMNet achieves state-of-the-art performance on the CDD-CD dataset with an $F1$-score while only requiring 6.94 G FLOPs.

For a more intuitive comparison, we present the visualization results of SPMNet against other methods in Fig. 7. As illustrated in Fig. 7(a), although the FLOPs of SPMNet are higher than those of FC-series models, it improves the $F1$-score by 35.19% over the FC-series models FC-Siam-Conv, which has the highest $F1$-score among such models. Furthermore, SPMNet's FLOPs are significantly lower than most attention-based models and even lower than SEIFNet, which focuses on lightweight design. Compared with other Mamba-based models of the same type, the FLOPs of ChangeMamba and RSM-CD are four times and twice that of SPMNet, respectively. This further demonstrates the fast inference capability of SPMamba (CNN–Mamba structure) in real-time tasks. However, it is worth noting that, as shown in Fig. 7(b), the number of parameters for SPMNet is at a moderate level. This is primarily because high-precision dense

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detection requires a more complex model structure. In summary, the SFPN architecture compensates for the shortcomings of the Mamba model in capturing complex texture features, allowing SPMamba to effectively balance contextual semantic information and focus more on change information. At the same time, the linear growth of the SSM enables SPMamba to achieve higher prediction accuracy with lower computational costs.

D. Ablation Study

To further validate the effectiveness of the proposed method, we conducted extensive ablation experiments and visual comparisons focusing on the Siamese pyramid structure, including the OSSM Block, SPMamba, HFM, and loss weights.

1) Ablation on OSSM Block: To validate that Mamba outperforms attention mechanisms and convolutional operations in complex texture feature extraction and analysis, we chose to replace OSSM with CBAM [51], SRCNN [52], and Swin Transformer [40] to evaluate the model performance. CBAM is a lightweight and efficient attention mechanism module that enhances the model's focus on key features by introducing channel attention and spatial attention. SRCNN, as a classic convolutional neural network, significantly improves the edges and texture details of images through detail restoration and super-resolution enhancement, thereby enhancing the model's ability to detect super-resolution remote sensing images. Swin Transformer, leveraging the self-attention mechanism, effectively models the long-range dependencies between pixels or regions in remote sensing images, making it suitable for multiscale change detection.

As shown in Table III, compared with SRCNN, SPMamba, which uses OSSM as its core, achieved a 7.09% improvement in $F1$-score on the WHU-CD dataset, with other metrics also leading comprehensively. This indicates that networks based on the CNN–Mamba architecture outperform pure CNN models. In addition, in the comparison of attention mechanisms, SPMNet, which is based on OSSM, significantly outperformed convolutional attention and self-attention mechanisms, achieving an improvement of 7.29% and 4.16% in $F1$-score on the LEVIR-CD dataset, respectively. Notably, on the CDD dataset, the FLOPs and parameter count of SRCNN and Swin Transformer are significantly higher than those of OSSM, while their $F1$-score decreased by 9.13% and 5.29%, respectively. This further validates the efficiency and superiority of SPMamba in VHR remote sensing image change detection tasks.

2) Ablation on SPMamba: As shown in Table IV, when the SPMamba structure is employed in the Backbone model, the precision on the WHU-CD dataset and the recall on the LEVIR-CD dataset improve by 1.79% and 3.55%, respectively. In addition, all evaluation metrics on the CDD-CD dataset show significant improvements. Compared with the Backbone, the model that employs SPMamba significantly reduces missed detections (red pixels) when identifying large buildings in the WHU-CD dataset and high-density, small-scale building clusters in the LEVIR-CD dataset. Similarly, it also reduces missed detections under vegetation change interference in the CDD-CD dataset. This improvement can be attributed to the combination of multiscale feature extraction and OSSM provided by SPMamba, enabling effective capture of complex texture features in the images.

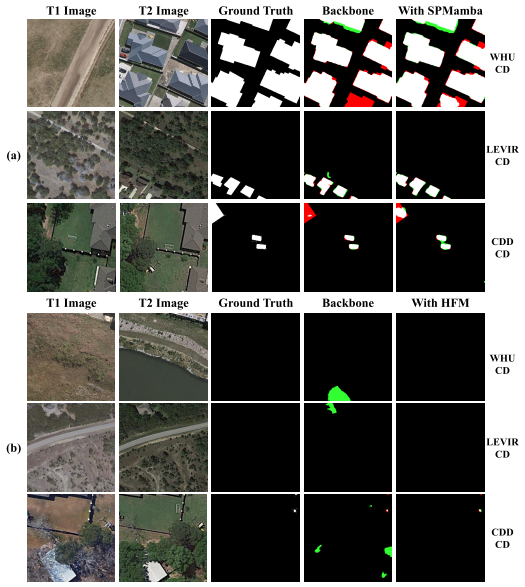


Fig. 8. Visualization of ablation study results on the WHU-CD, LEVIR-CD, and CDD-CD datasets. (a) Ablation of SPMamba. (b) Ablation of HFM.

3) Ablation on HFM: As shown in Table IV, when using HFM alone, all metrics across the three datasets improve. Notably, on the CDD-CD dataset, precision, recall, and $F1$-score increase by 2.22%, 5.13%, and 3.69%, respectively. Fig. 8(b) shows this effect, where the backbone without HFM mistakenly identifies unchanged pixels as changed pixels when processing nonchange pixel data in WHU-CD and CDD-CD datasets. Similarly, strong lighting conditions affect the detection task on LEVIR-CD dataset, leading to false detections. The analysis reveals that HFM first utilizes the channel-level fusion HLM to eliminate potential noise in multiscale semantic features, followed by the application of the feature-level fusion BCM to enhance semantic recognition, improving the model's sensitivity to both changed and unchanged pixels. In summary, this approach allows the model to acquire more accurate high-level features from minimal foreground information, thereby reducing misjudgments and false change detections.

4) Ablation on Loss: For the change detection task with complex texture features and extreme foreground–background imbalance, we adopted a combination of BCE and Dice loss functions. BCE loss emphasizes pixel-level detail classification, while Dice loss focuses on the overall consistency of the foreground regions. The combination of both can optimize local details and global region consistency. Therefore, finding the best loss function combination can effectively improve the change detection performance. As shown in Table V, we compared the performance of three different weight combinations on WHU-CD dataset. Compared with the combined loss with the weight combination ($\lambda = 1.0$ and $\Phi = 1.0$), when the weight combination was ($\lambda = 0.3$ and $\Phi = 0.7$), the overall performance of SPMNet decreased ($F1$-score decreased by 3.29%). This indicates that the model is overly focused on foreground consistency, neglecting local detail features. When

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TABLE III

ABLATION EXPERIMENT RESULTS(%) ON THE CORE PROCESSING COMPONENTS OF SPMAMBA ON WHU-CD, LEVIR-CD, AND CDD-CD DATASETS. THE VALUES IN BOLD ARE THE BEST

Methods	FLOPs(G)	Params(M)	WHU-CD			LEVIR-CD			CDD-CD		
			Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
CBAM [51]	1.25	4.83	92.95	75.57	83.36	88.65	81.30	84.81	91.78	88.15	89.93
SRCNN [52]	40.48	31.99	94.33	78.54	85.72	91.39	87.45	89.38	90.46	84.28	87.27
SwinT [40]	12.90	27.65	92.60	82.20	87.09	88.59	86.17	87.36	92.75	89.24	90.96
OSSM	6.94	17.33	**94.67**	**89.10**	**91.80**	**92.12**	**90.58**	**90.99**	**96.63**	**95.60**	**96.04**

TABLE IV

ABLATION EXPERIMENT RESULTS(%) ON WHU-CD, LEVIR-CD, AND CDD-CD DATASETS. THE VALUES IN BOLD ARE THE BEST

Methods	SPMamba	HFM	WHU-CD			LEVIR-CD			CDD-CD		
			Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
Backbone	✗	✗	92.68	85.34	89.05	91.07	86.78	89.63	94.04	90.17	92.06
Backbone	✓	✗	94.34	85.61	89.76	91.51	89.86	90.46	94.39	91.98	93.17
Backbone	✗	✓	93.49	85.48	89.31	91.34	90.24	90.79	96.13	94.80	95.46
Backbone	✓	✓	**94.67**	**89.10**	**91.80**	**92.12**	**90.58**	**90.99**	**96.63**	**95.60**	**96.04**

TABLE V

ABLATION EXPERIMENT RESULTS (%) ON THE LOSS FUNCTION PARAMS OF THE WHU-CD DATASET. THE VALUES IN BOLD ARE THE BEST

Loss Parameter	Pre.	Rec.	F1	IoU	OA.
$\lambda = 0.3, \Phi = 0.7$	93.88	84.20	88.78	79.82	99.09
$\lambda = 0.7, \Phi = 0.3$	94.52	78.39	85.70	74.98	98.88
$\lambda = 1.0, \Phi = 1.0$	**94.67**	**89.10**	**91.80**	**84.84**	**99.32**

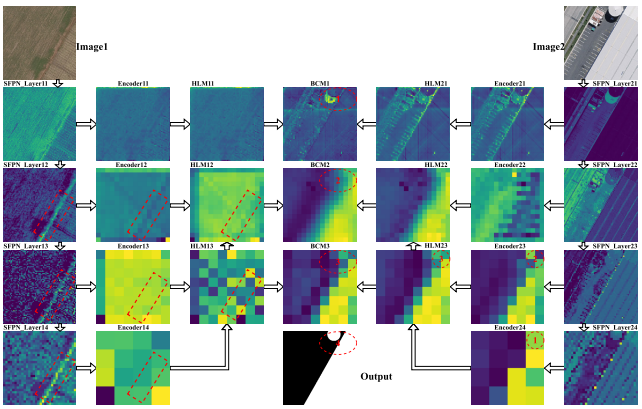


Fig. 9. Visualization results represent the multiscale feature maps of the main stages based on the SPMNet structure, including the SFPN, encoder (based on OSSM), HLM, and BCM stages. Image 1 and Image 2 denote the input bitemporal images, while the output represents the final detection result map.

the weight combination was ($\lambda = 0.7$ and $\Phi = 0.3$), with BCE loss dominating, SPMNet focused more on background elements, leading to a 12.02% drop in recall. In summary, only when BCE and Dice losses are balanced can SPMNet achieve the best detection performance in highly imbalanced remote sensing change detection tasks.

E. Visualization Analysis of SPMNet

To better illustrate the performance improvements achieved by SPMNet and validate the effectiveness of the proposed multiscale feature extraction and fusion method, we averaged the feature maps at each stage across the channel dimension to generate composite feature maps that reflect overall characteristics. Specifically, as shown in Fig. 9, we conducted a visual analysis of the feature maps at each stage of the SFPN, encoder, and HFM according to the structure of the SPMNet model diagram. It is important to note that brighter elements represent higher scores and are more likely to be classified as change regions, with key focus areas marked by red dashed boxes.

As illustrated in Fig. 9, Image 1 and Image 2 first undergo bottom-up multiscale feature extraction through the SFPN, providing multiscale local salient feature information for the OSSM-based encoder. On this basis, different levels of the encoder further capture global features, achieving the unification of local and global feature representations. From the SFPN feature maps marked with rectangular boxes in Fig. 9, it can be seen that simple multiscale convolutional operations (SFPN12, 13, 14) mistakenly regard surface subsidence as a change element. However, after spatial feature capture by the corresponding encoder layers (Encoders 12–14), the feature maps align more closely with the original images. For the feature maps marked with circles in Fig. 9, there are significant differences in the circular regions of Encoder 23 and Encoder 24, while the corresponding circular regions in HLM23 are more consistent with the output. From the feature maps marked with elliptical regions in Fig. 9, it is evident that after BCM fusion, the interference of noisy features is reduced.

Overall, SPMNet effectively preserves rich local features while enhancing global feature capture and shallow feature representation. This allows SPMNet to better handle complex ground objects of different scales.

F. Impact of Different Image Sizes

The size and spatial context information of very high-resolution remote sensing images are crucial for change detection tasks. The size of the image determines the level of detail that can be observed, aiding in the identification of subtle changes and large-scale change areas, while also

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TABLE VI
QUANTITATIVE COMPARISON OF DIFFERENT IMAGE SIZES ON THE WHU-CD AND LEVIR-CD DATASETS. THE VALUES IN BOLD ARE THE BEST

Datasets	Resolution	Pixels(M)	Method	Image Size			Pre.	Rec.	F1	IoU	OA.	Params(M)	FLOPs(G)
				256	512	1024							
WHU CD	0.3m	503.32	SPMNet	✓			94.67	**89.10**	**91.80**	**84.84**	**99.32**	17.33	**6.94**
					✓		94.51	84.54	89.25	80.58	99.18	17.33	27.79
						✓	**94.73**	72.30	82.01	69.50	98.72	17.33	111.19
LEVIR CD	0.5m	667.94	SPMNet	✓			92.12	**90.58**	**90.99**	**84.07**	**99.09**	17.33	**6.94**
					✓		**92.45**	88.13	90.24	82.21	99.02	17.33	27.79
						✓	89.79	86.17	87.94	78.48	98.80	17.33	111.19

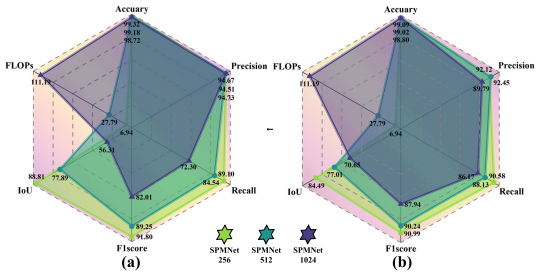


Fig. 10. Visualization of image sizes on the WHU-CD and LEVIR-CD datasets. (a) WHU-CD Dataset. (b) LEVIR-CD Dataset.

allowing for multiscale analysis to capture features at different scales. Recently, Talebi and Milanfar [53] proposed a learnable resizer model to find the optimal image size for model training by comparing the results of multiple downsampling methods. In addition, Zhao et al. [18] applied multiscale downsampling through bilinear interpolation to analyze the impact of VHR remote sensing images on model training. Therefore, we proposed three training strategies: SPMNet1024, SPMNet512, and SPMNet256. Specifically, we cropped the WHU-CD and LEVIR-CD datasets, with total pixel sizes equivalent to 455.08 and 667.94 m, respectively, into nonoverlapping 1024 × 1024 images (SPMNet1024). We further divided these images into nonoverlapping 512 × 512 (SPMNet512) and 256 × 256 (SPMNet256) images. It is important to note that the change instances in the WHU-CD and LEVIR-CD datasets are exclusively buildings, which ensures consistency in the overall change semantic features during model training.

Table VI presents the final testing results for images with sizes of 256 × 256, 512 × 512, and 1024 × 1024 pixels. The results indicate that, with a fixed number of detection instances (total pixel count), the evaluation metrics generally improve as the image size decreases. Specifically, on the WHU-CD dataset, SPMNet256 shows improvements in recall by 5.39% and 23.24%, and in IoU by 5.29% and 22.07% compared with SPMNet512 and SPMNet1024, respectively. On the LEVIR-CD dataset, when cropping the image size to 256 × 256, the IoU improves by 7.12% compared with SPMNet1024. This improvement is primarily due to the fact that larger remote sensing images contain a large amount of background area and only a small portion of true positives. The presence of background noise, including textures or color features similar to the target change areas, leads to an increase in misdetections.

As shown in Fig. 10, the visual comparison results of different training strategies on the WHU-CD and LEVIR-CD datasets are displayed. Fig. 10(a) and (b) indicates that the overall performance of the model is inversely proportional to the image size, with a significant performance improvement observed when the image size decreases from 1024 × 1024 to 512 × 512. In addition, we evaluated the model's Params and FLOPs. Due to the SSM mechanism, the FLOPs of SPMNet decrease linearly with the increase in image size, from a maximum of 111.19 G down to 6.94 G. The visual results further show that, while larger images contain more spatial context information, they inevitably include a large amount of scattered redundant information. This dispersion of features can cause the model to lose focus on the change areas, ultimately reducing both the precision and efficiency of the model's detection.

V. CONCLUSION

To investigate the complex texture features and background noise of VHR bitemporal remote sensing images, we propose a deep learning method called SPMNet and introduce the pyramid-based Mamba for the first time in change detection tasks. First, SPMNet utilizes Mamba as a core component and integrates it with SFPN to construct the SPMamba, modeling multiscale and multidirectional image features while maintaining linear computational complexity. In addition, we design a novel fusion module, HFM, which combines multiscale spatial features and interdomain differences across multiple stages, leading to more comprehensive and accurate change detection results. Through both quantitative and qualitative analyses on four well-recognized change detection datasets, our proposed SPMNet outperforms existing advanced change detection methods. In future research, we aim to further improve and optimize the performance of the network and apply it to multitemporal change detection tasks. We also hope that our work will inspire additional research into the integration of convolutional neural networks and Mamba to enhance the interpretation of bitemporal remote sensing images.

REFERENCES

[1] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.
[2] P. K. Mishra, A. Rai, and S. C. Rai, "land use and land cover change detection using geospatial techniques in the Sikkim Himalaya, India," *Egyptian J. Remote Sens. Space Sci.*, vol. 23, no. 2, pp. 133–143, Aug. 2020.
[3] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:27:13 UTC from IEEE Xplore. Restrictions apply.

[4] S. Asadzadeh, W. J. D. Oliveira, and C. R. D. Souza Filho, "UAV-based remote sensing for the petroleum industry and environmental monitoring: State-of-the-art and perspectives," *J. Petroleum Sci. Eng.*, vol. 208, Jan. 2022, Art. no. 109633.

[5] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[6] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[7] R. C. Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Urban change detection for multispectral Earth observation using convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 2115–2118.

[8] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2018.

[9] A. Varghese, J. Gubbi, A. Ramaswamy, and P. Balamuralidhar, "ChangeNet: A deep learning architecture for visual change detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2019, pp. 129–145.

[10] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[11] W. Shi, M. Zhang, H. Ke, X. Fang, Z. Zhan, and S. Chen, "Landslide recognition by deep convolutional neural network and change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4654–4672, Jun. 2020.

[12] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[13] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 30, Jun. 2017, pp. 5998–6008.

[14] Y. Li, H. Chen, C. Kong, L. Dai, and X. Chen, "Learning an anchor-free network with angle attention for remote sensing object detection," *Remote Sens. Lett.*, vol. 14, no. 9, pp. 936–946, Sep. 2023.

[15] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[16] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[17] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[18] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," 2024, *arXiv:2404.02668*.

[19] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[20] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019, doi: 10.1109/TGRS.2018.2858817.

[21] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[22] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 565–571, May 2018.

[23] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067, doi: 10.1109/ICIP.2018.8451652.

[24] Z. Zhang, G. Vosselman, M. Gerke, D. Tuia, and M. Y. Yang, "Change detection between multimodal remote sensing data using Siamese CNN," 2018, *arXiv:1807.09562*.

[25] G. Shi, J. Zhang, J. Liu, C. Zhang, C. Zhou, and S. Yang, "Global context-augmented objection detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10604–10617, Dec. 2021.

[26] L. Ding, H. Tang, Y. Liu, Y. Shi, X. X. Zhu, and L. Bruzzone, "Adversarial shape learning for building extraction in VHR remote sensing images," *IEEE Trans. Image Process.*, vol. 31, pp. 678–690, 2022.

[27] J. Jiang, J. Xiang, E. Yan, Y. Song, and D. Mo, "Forest-CD: Forest change detection network based on VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] Z. Chen, Y. Song, Y. Ma, G. Li, R. Wang, and H. Hu, "Interaction in transformer for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3000612.

[29] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jul. 2022, Art. no. 5607514, doi: 10.1109/TGRS.2021.3095166.

[30] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519, doi: 10.1109/TGRS.2022.3169479.

[31] A. Gu, T. Dao, S. Ermon, A. Rudra, and C. Ré, "HiPPO: Recurrent memory with optimal polynomial projections," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jan. 2020, pp. 1474–1487.

[32] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 34, 2021, pp. 572–585.

[33] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[34] A. Gu, A. Gupta, K. Goel, and C. Ré, "On the parameterization and initialization of diagonal state space models," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jan. 2022, pp. 35971–35983.

[35] A. Gupta, "Diagonal state spaces are as effective as structured state spaces," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Jan. 2022, pp. 22982–22994.

[36] J. N. Paranjape, C. de Melo, and V. M. Patel, "A Mamba-based Siamese network for remote sensing change detection," 2024, *arXiv:2407.06839*.

[37] J. Zhang, R. Chen, F. Liu, H. Liu, B. Zheng, and C. Hu, "DC-Mamba: A novel network for enhanced remote sensing change detection in difficult cases," *Remote Sens.*, vol. 16, no. 22, p. 4186, Nov. 2024.

[38] Y. Liu, G. Cheng, Q. Sun, C. Tian, and L. Wang, "CWmamba: Leveraging CNN-Mamba fusion for enhanced change detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[39] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[40] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[41] L. Dai et al., "TARDet: Two-stage anchor-free rotating object detector in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 4266–4274.

[42] Y.-D. Ma, Q. Liu, and Z.-B. Quan, "Automated image segmentation using improved PCNN model based on cross-entropy," in *Proc. Int. Symp. Intell. Multimedia, Video Speech Process.*, Oct. 2004, pp. 743–746.

[43] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. J. Cardoso, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Proc. DLMIA ML-CDS*, 2017, pp. 240–248.

[44] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.

[45] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021, doi: 10.1109/LGRS.2020.2988032.

[46] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2021.3056416.

[47] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210, doi: 10.1109/IGARSS46834.2022.9883686.

[48] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023, doi: 10.1109/JSTARS.2023.3264802.

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[49] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414, doi: 10.1109/TGRS.2024.3360516.

[50] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720, doi: 10.1109/TGRS.2024.3417253.

[51] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[52] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[53] H. Talebi and P. Milanfar, "Learning to resize images for computer vision tasks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 497–506.



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